28/9/24

EX NO 7

Practical - 7

Flow control and Data link layer

Aim:

Write a program to implement flow control at data link layer using sliding window protocol

Sender program

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#define MAX 100

struct frame {
        int frame_no;
        char data;
};
int main() {
    struct frame frames[MAX];
    int windowsize, n, i, start=0;
    char message[MAX];
    file fp;
    int ack;
    scanf("%d", &windowsize);
    scanf("%s", &message);
    n= strlen(message);
    for(int i=0; i<n; i++) {
        frames[i].frame_no = i+1;
        frames[i].data = message[i];
    while(start<n) {
        int end = start + windowsize;
        if(end>n) end=n;
```

Construct a CFG for the Language L over such that $w \in (a+b)^*$

```
fp = fopen("sender_Buffer", "w");
if (fp == NULL) {
        printf("error");
        return 1;
}
for (i = start ; i <= end ; i++) {
        printf(fp, "%d' %c \n", frames[i].frame_no,
                frames[i].data );
        printf("frameNo : %d DATA: %c \n",
                frames[i].frame_no, frames[i].data);
}
fclose(fp);
printf("Frames stored in sender Buffer");
fp = fopen("RecieverBuffer", "r");
if (fp == NULL) {
        continue;
}
fscanf(fp, "%d", &ack);
fclose(fp);
printf("Recieved ACK/NACK %d\n", ack);
if (ack >= start+1 && ack <= end) {
        printf("ACK recieved", ack);
        start = ack;
        printf("Sliding window moved forward \n");
} else if (ack < 0) {
        ack = -ack;
        printf("NACK recieved for frame %d\n"
                ack);
}
printf("All frames transmitted successfully");
return 0;
}
```

Reciever program

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
int main() {
    FILE *fp, *out;
    int frameNo;
    char data;
    int expected = 1;
    while (1) {
        fp = fopen("sender_Buffer", "r");
        if (fp == NULL) {
            printf("waiting");
            sleep(2);
            continue;
        }
        while fscanf(fp, "%d", %c", & frame_no, &data)
        if (frame_no == expected) {
            out = fopen("Reciever_Buffer", "w");
            if (out == NULL) {
                fclose(fp);
                return 0;
            }
        }
        fclose(fp);
        return 0;
    }
```

Result:

The code has been executed successfully